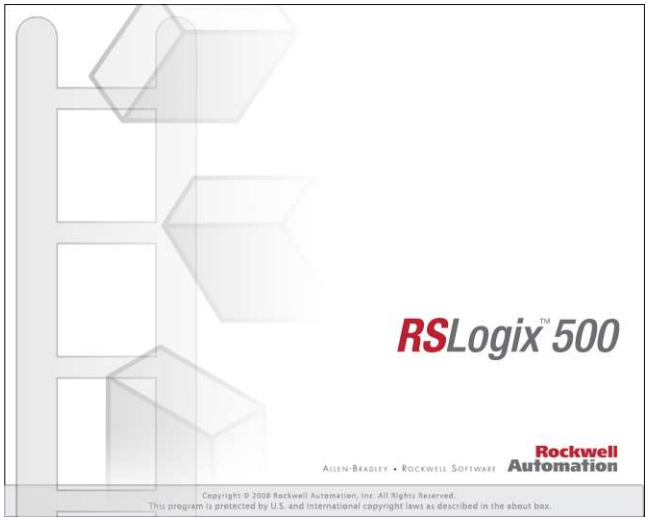
RSLogix 500 Project Report



Processor Information

Processor Type: Bul.1762 MicroLogix 1200 Series C (1 or 2 Comm Ports)

Processor Name: UNTITLED

Total Memory Used: 493 Instruction Words Used - 107 Data Table Words Used

Total Memory Left: 5203 Instruction Words Left

Program Files: 6

Data Files: 9

Program ID: 259e

I/O Configuration

0	Bul.1762	MicroLogix 1200 Series C (1 or 2 C
1		
2		
3		
4		
5		
6		

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
   CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

Prog/HMI Port - Driver: DF1 Full Duplex
   Source ID: 1 (decimal)
   Baud: 19200
   Parity: NONE
   Control Line : No Handshaking
   Error Detection: CRC
   Embedded Responses: Auto Detect
   Duplicate Packet Detect: Yes
   ACK Timeout(x20 ms): 50
   NAK Retries: 3
   ENQ Retries: 3

Program File List

Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
	2	LADDER	35	No	1042
SUB0	3	LADDER	23	No	712
SUB1	4	LADDER	13	No	282
SUB2	5	LADDER	10	No	172

Data File List

Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	12	4	O:3
INPUT	1	I	Global	No	12	4	I:3
STATUS	2	S	Global	No	0	66	S:65
BINARY	3	B	Global	No	5	5	B3:4
TIMER	4	T	Global	No	60	20	T4:19
COUNTER	5	C	Global	No	12	4	C5:3
CONTROL	6	R	Global	No	3	1	R6:0
INTEGER	7	N	Global	No	1	1	N7:0
FLOAT	8	F	Global	No	2	1	F8:0

(MAIN) STOP/PAUSE SEQUENCE

if STOP is pressed

0000

EQU
Equal
Source A I:0.0 36<
Source B 32 32<

DRILL INDICATION
O:0
U
1
Bul.1762

CLAMP EXTEND INDICATION
O:0
U
3
Bul.1762

PAUSE MEMORY
B3:0
L
10

if START button is pressed and the SELECTOR is on AUTO (=0)

0001

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC 3<
Source B 0 0<

EQU
Equal
Source A I:0.0 36<
Source B 37 37<

PAUSE MEMORY
B3:0
U
10

filter out unwanted press on SELECTOR

0002

EQU
Equal
Source A I:0.0 36<
Source B 44 44<

INPUT_DELAY_TIMER
TON
Timer On Delay
Timer T4:18
Time Base 0.01
Preset 10<
Accum 0<
EN
DN

0003

INPUT_DELAY_TIMER/DN
T4:18
DN

SELECTOR COUNTER
SELECTOR_1
CTU
Count Up
Counter C5:0
Preset 5<
Accum 3<
CU
DN

MOV
Move
Source 0 0<
Dest O:0.0 0<

INITIALISE OUTPUT TO ZERO (ALL TURNED OFF); reset to zero the selection counter

SELECTOR COUNTER
SELECTOR_1.ACC

0004

EQU
Equal
Source A C5:0.ACC
3<
Source B 4
4<

MOV
Move
Source 0
0<
Dest O:0.0
0<

CONVEYOR RUNNING .8 DELAY
T4:6
RES

1 SEC DELAY FOR CLAMP EXTENDING
T4:7
RES

1 SEC DELAY FOR CLAMP EXTENDING
T4:7
RES

2 SECOND DELAY FOR DRILL STARTING
T4:8
RES

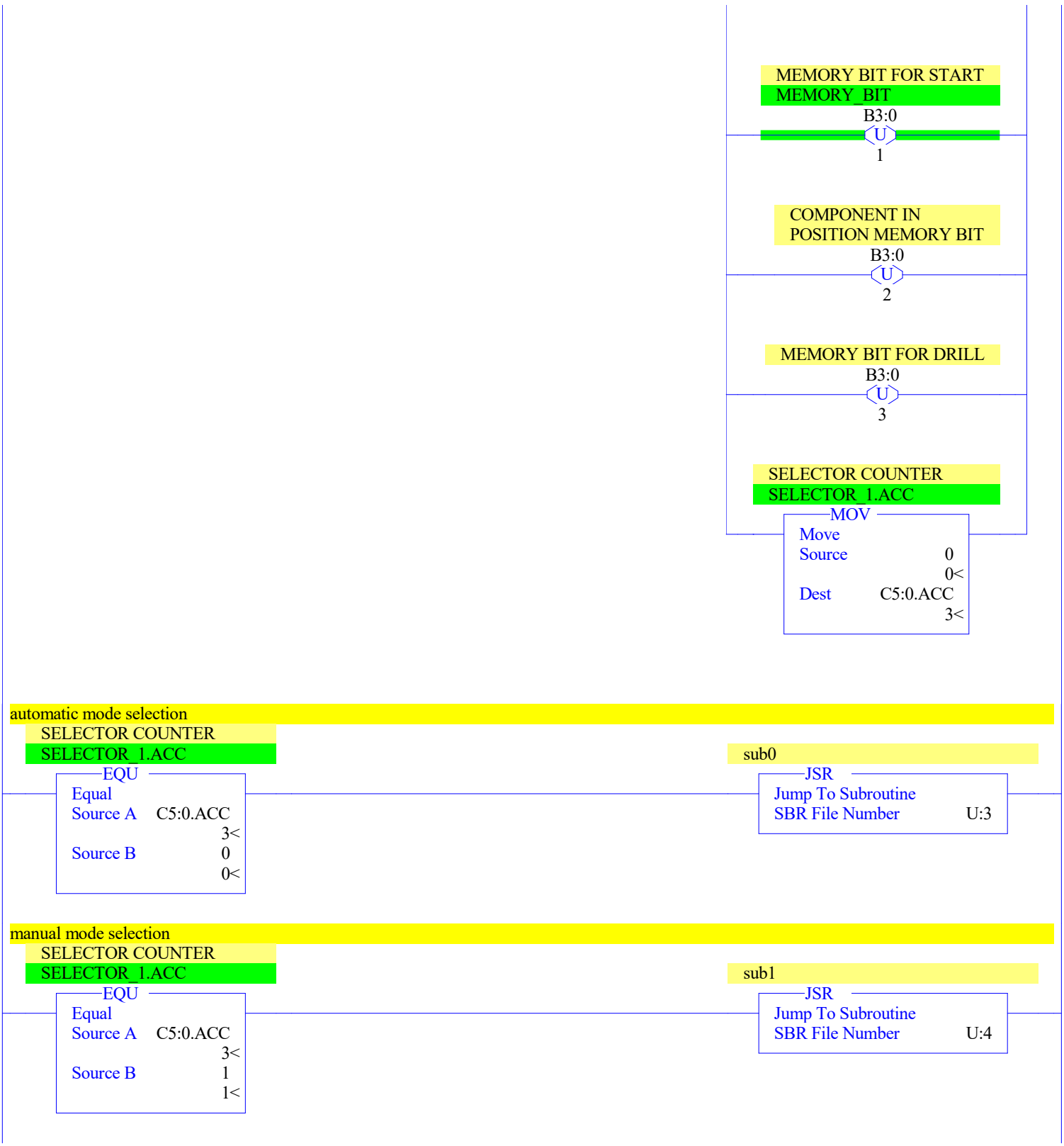
FLASHING CIRCUIT T1
T4:9
RES

FLASHING CIRCUIT T2
T4:10
RES

5 SECOND TIMER FOR DRILL RUNNING
T4:11
RES

TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR
T4:12
RES

TIMER TO RESET COUNTER IF INPUTS TAKE LONGER THAN 2SEC
T4:13
RES


MEMORY BIT FOR START
MEMORY_BIT
B3:0
U
1
COMPONENT IN POSITION MEMORY BIT
B3:0
U
2
MEMORY BIT FOR DRILL
B3:0
U
3
SELECTOR COUNTER
SELECTOR_1.ACC
MOV
Move
Source 0
0<
Dest C5:0.ACC
3<
automatic mode selection
SELECTOR COUNTER
SELECTOR_1.ACC
EQU
Equal
Source A C5:0.ACC
3<
Source B 0
0<
0005
sub0
JSR
Jump To Subroutine
SBR File Number U:3
manual mode selection
SELECTOR COUNTER
SELECTOR_1.ACC
EQU
Equal
Source A C5:0.ACC
3<
Source B 1
1<
0006
sub1
JSR
Jump To Subroutine
SBR File Number U:4

Eng Mode catching

enable lights on pbuttons only for the eng mode;;;
no press of any button

0007

FAULTY COMPONENT
I:0
5
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

test
B3:0
13

0008

STOP BUTTON
I:0
2
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

B3:0
14

0009

START BUTTON
I:0
0
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

B3:0
15

0010

MANUAL BUTTON
I:0
1
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

B3:4
0

0011

INPUT SELECTOR
SELECTOR
I:0
3
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

B3:4
1

0012

COMPONENT IN POSITION
I:0
4
Bul.1762

SELECTOR COUNTER
SELECTOR_1.ACC

EQU
Equal
Source A C5:0.ACC
3<
Source B 2
2<

B3:4
2

engineering secret code sequence entry

0013

EQU
Equal
Source A I:0.0
36<
Source B 38
38<

state A
B3:0
L
6

0014

state A
B3:0
6

EQU
Equal
Source A I:0.0
36<
Source B 14
14<

state B
B3:0
L
8

0015

state B
B3:0
8

EQU
Equal
Source A I:0.0
36<
Source B 49
49<

state C
B3:0
L
9

0016

state C
B3:0
9

EQU
Equal
Source A I:0.0
36<
Source B 28
28<

state D
B3:0
L
11

0017

state D
B3:0
11

EQU
Equal
Source A I:0.0
36<
Source B 35
35<

ENG MODE ON
B3:0
L
12

state D
B3:0
U
11

state C
B3:0
U
9

state B
B3:0
U
8

state A
B3:0
U
6

0018

ENG MODE ON
B3:0
12

sub2

JSR
Jump To Subroutine
SBR File Number U:5

STUCK INPUTS

FAULTY COMPONENT

0019

I:0
5
Bul.1762

TON
Timer On Delay
Timer T4:0
Time Base 1.0
Preset 10<
Accum 0<
EN
DN

COMPONENT IN POSITION

0020

I:0
4
Bul.1762

TON
Timer On Delay
Timer T4:3
Time Base 1.0
Preset 10<
Accum 0<
EN
DN

START BUTTON

0021

I:0
0
Bul.1762

TON
Timer On Delay
Timer T4:4
Time Base 1.0
Preset 10<
Accum 0<
EN
DN

STOP BUTTON

0022

I:0
2
Bul.1762

TON
Timer On Delay
Timer T4:5
Time Base 1.0
Preset 10<
Accum 0<
EN
DN

FLASHING SIGNAL (SECT 1)

0023

T4:2
DN

TON
Timer On Delay
Timer T4:1
Time Base 0.01
Preset 50<
Accum 50<
EN
DN

FLASHING SIGNAL, SEC. 2 ; PEAKS

0024

T4:1
DN

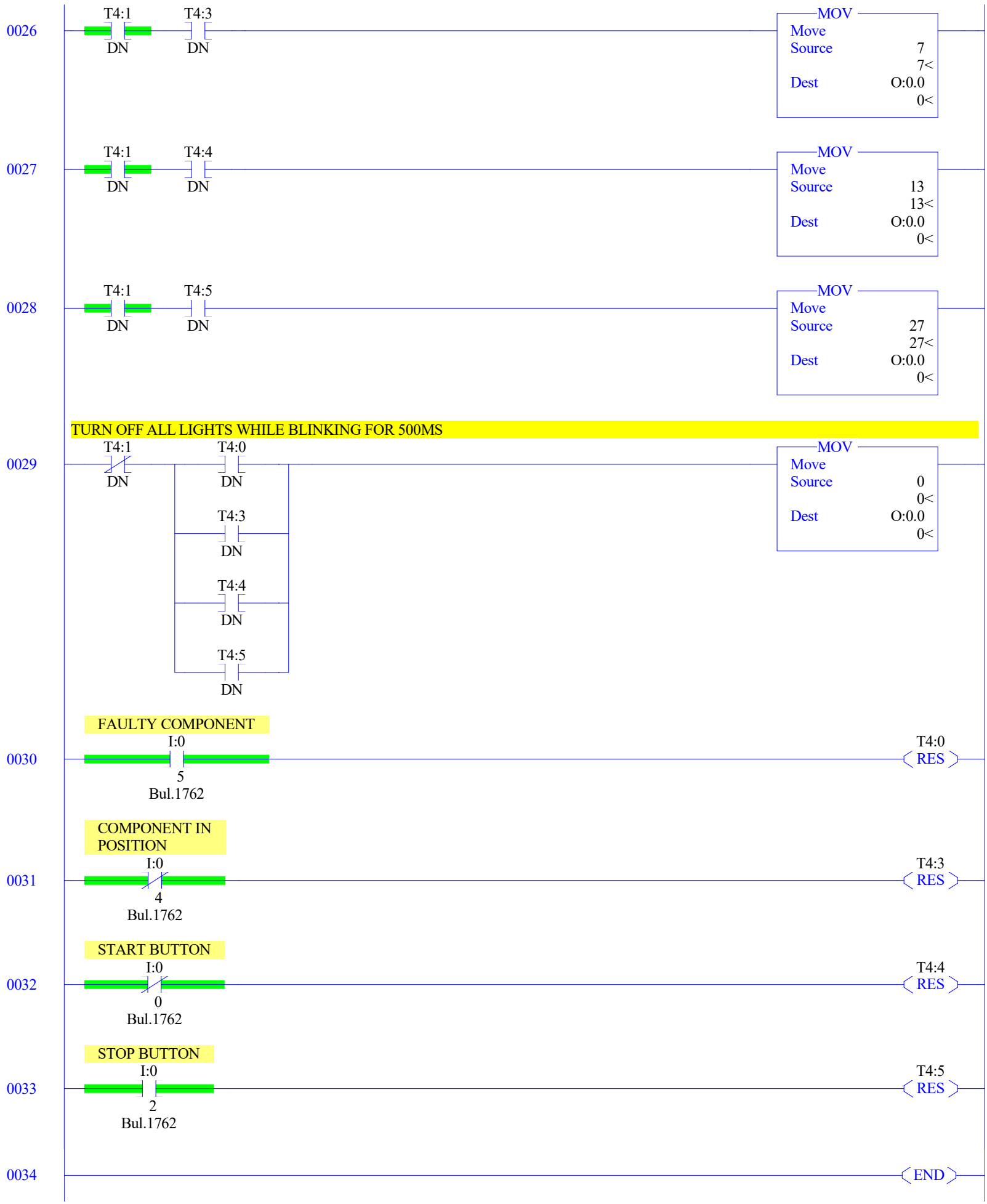
TON
Timer On Delay
Timer T4:2
Time Base 0.01
Preset 50<
Accum 3<
EN
DN

0025

T4:1
DN

T4:0
DN

MOV
Move
Source 3
3<
Dest O:0.0
0<


0026
T4:1
DN
T4:3
DN
MOV
Move
Source 7
7<
Dest O:0.0
0<
0027
T4:1
DN
T4:4
DN
MOV
Move
Source 13
13<
Dest O:0.0
0<
0028
T4:1
DN
T4:5
DN
MOV
Move
Source 27
27<
Dest O:0.0
0<
TURN OFF ALL LIGHTS WHILE BLINKING FOR 500MS
0029
T4:1
DN
T4:0
DN
T4:3
DN
T4:4
DN
T4:5
DN
MOV
Move
Source 0
0<
Dest O:0.0
0<
FAULTY COMPONENT
0030
I:0
5
Bul.1762
T4:0
RES
COMPONENT IN POSITION
0031
I:0
4
Bul.1762
T4:3
RES
START BUTTON
0032
I:0
0
Bul.1762
T4:4
RES
STOP BUTTON
0033
I:0
2
Bul.1762
T4:5
RES
0034
END

filters out unwanter press on START button

0000

EQU
Equal
Source A I:0.0
36<
Source B 37
37<

TON
Timer On Delay
Timer T4:19
Time Base 0.01
Preset 5<
Accum 0<
EN
DN

INITIALISE ALL LIGHTS TO OFF (TODO)

0001

T4:19
DN

MEMORY BIT FOR START
MEMORY_BIT
B3:0
L
1

TIMING SIGNAL T_ON; PART 1

0002

FLASHING CIRCUIT T2
T4:10
DN

FLASHING CIRCUIT T1
TON
Timer On Delay
Timer T4:9
Time Base 0.01
Preset 20<
Accum 20<
EN
DN

TIMING SIGNAL PEAKS: PART 2

0003

FLASHING CIRCUIT T1
T4:9
DN

5 SECOND TIMER FOR DRILL RUNNING
T4:11
DN

FLASHING CIRCUIT T2
TON
Timer On Delay
Timer T4:10
Time Base 0.01
Preset 20<
Accum 13<
EN
DN

0004

MEMORY BIT FOR START
MEMORY_BIT
B3:0
1

CONVEYOR RUN
O:0
0
Bul.1762

CLAMP RETRACTED LIGHT
O:0
2
Bul.1762

if SELECTOR counter is =0, DELAY TO STOP THE CONVEYOR (0.8SEC)

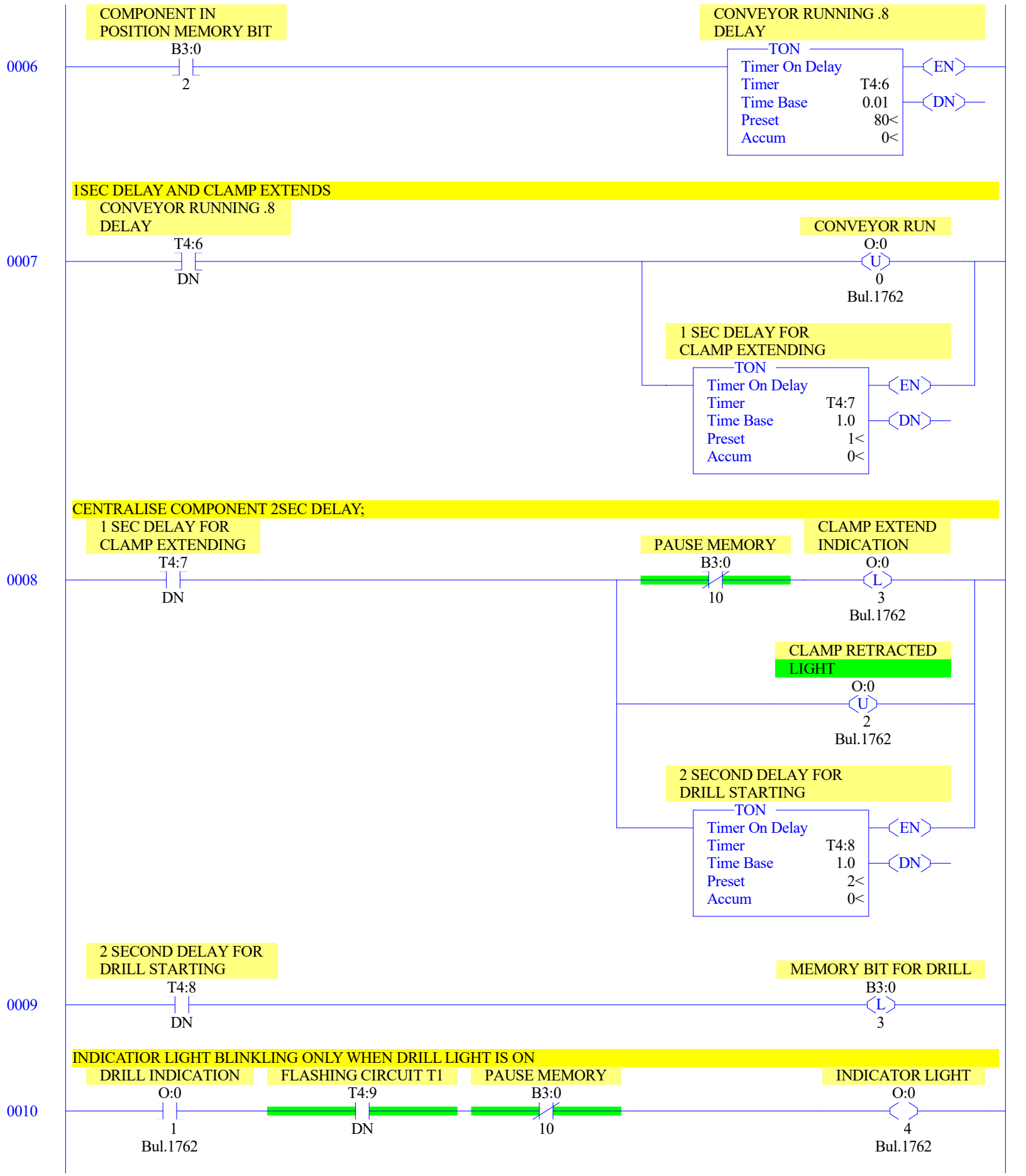
0005

SELECTOR COUNTER
SELECTOR_1.ACC
EQU
Equal
Source A C5:0.ACC
3<
Source B 0
0<

COMPONENT IN POSITION
I:0
4
Bul.1762

PAUSE MEMORY
B3:0
10

COMPONENT IN POSITION MEMORY BIT
B3:0
L
2


COMPONENT IN POSITION MEMORY BIT
B3:0
2
CONVEYOR RUNNING .8 DELAY
TON
Timer On Delay
Timer T4:6
Time Base 0.01
Preset 80<
Accum 0<
EN
DN
0006
1SEC DELAY AND CLAMP EXTENDS
CONVEYOR RUNNING .8 DELAY
T4:6
DN
CONVEYOR RUN
O:0
U
0
Bul.1762
1 SEC DELAY FOR CLAMP EXTENDING
TON
Timer On Delay
Timer T4:7
Time Base 1.0
Preset 1<
Accum 0<
EN
DN
0007
CENTRALISE COMPONENT 2SEC DELAY;
1 SEC DELAY FOR CLAMP EXTENDING
T4:7
DN
PAUSE MEMORY
B3:0
10
CLAMP EXTEND INDICATION
O:0
L
3
Bul.1762
CLAMP RETRACTED LIGHT
O:0
U
2
Bul.1762
2 SECOND DELAY FOR DRILL STARTING
TON
Timer On Delay
Timer T4:8
Time Base 1.0
Preset 2<
Accum 0<
EN
DN
0008
2 SECOND DELAY FOR DRILL STARTING
T4:8
DN
MEMORY BIT FOR DRILL
B3:0
L
3
0009
INDICATIOR LIGHT BLINKLING ONLY WHEN DRILL LIGHT IS ON
DRILL INDICATION
O:0
1
Bul.1762
FLASHING CIRCUIT T1
T4:9
DN
PAUSE MEMORY
B3:0
10
INDICATOR LIGHT
O:0
4
Bul.1762
0010

START DRILLING FOR 5SEC AND MAKE IND.LIGHT BLINKING 0.2S/0.2S

0011

MEMORY BIT FOR DRILL
B3:0
3

DRILL INDICATION
O:0
L
1
Bul.1762

5 SECOND TIMER FOR DRILL RUNNING
TON
Timer On Delay
Timer T4:11
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

WAIT 2SEC AFTER DRILL STOPS

0012

5 SECOND TIMER FOR DRILL RUNNING
T4:11
DN

DRILL INDICATION
O:0
U
1
Bul.1762

TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR
TON
Timer On Delay
Timer T4:12
Time Base 1.0
Preset 2<
Accum 0<
EN
DN

MEMORY BIT FOR DRILL
B3:0
U
3

RETRACT CLAMP AND START CONVEYOR; TURN OFF CLAMP EXTEND AND TURN ON CLAMP RETRACT LIGHT

0013

TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR
T4:12
DN

PAUSE MEMORY
B3:0
10

CLAMP RETRACTED LIGHT
O:0
L
2
Bul.1762

CONVEYOR RUN
O:0
L
0
Bul.1762

CLAMP EXTEND INDICATION
O:0
U
3
Bul.1762

if SELECTOR counter =0; FAULT DETECTION: FAULT INPUT FOUR TIMES IN TWO SECONDS WILL TRIGGER; MAYBE USE MB HERE

0014

FAULTY COMPONENT
I:0
5
Bul.1762

FAULTY INPUT COUNTER1
CTU
Count Up
Counter C5:1
Preset 4<
Accum 0<
CU
DN

MEMORY BIT TO START TIMER FROM COUNTER INPUT
B3:0
L
4

0015

MEMORY BIT TO START TIMER FROM COUNTER INPUT
B3:0
4

TIMER TO RESET COUNTER IF INPUTS TAKE LONGER THAN 2SEC
TON
Timer On Delay
Timer T4:13
Time Base 1.0
Preset 2<
Accum 0<
EN
DN

PART OF FAULT DETECTION: RESET COUNTER AFTER TWO SECONDS

0016

TIMER TO RESET COUNTER IF INPUTS TAKE LONGER THAN 2SEC
T4:13
DN

FAULTY INPUT COUNTER1
C5:1
RES

FAULTY INPUT COUNTER 2
C5:2
RES

MEMORY BIT TO START TIMER FROM COUNTER INPUT
B3:0
U
4

0017

FAULTY INPUT COUNTER1
C5:1
DN

MEMORY BIT FOR LIGHT OUTPUTS
B3:0
L
5

INPUT STOPS CONVEYOR AND BRINGS ON FAULTY COMPONENT LIGHT FOR TWO SECONDS

0018

MEMORY BIT FOR LIGHT OUTPUTS
B3:0
5

FAULTY COMPONENT
O:0
L
5
Bul.1762

CONVEYOR RUN
O:0
U
0
Bul.1762

CLAMP RETRACTED LIGHT
O:0
U
2
Bul.1762

TIMER FOR FAULTY COMPONENT LIGHT
TON
Timer On Delay
Timer T4:14
Time Base 1.0
Preset 2<
Accum 0<
EN
DN

0019

TIMER FOR FAULTY COMPONENT LIGHT
T4:14
DN

CONVEYOR RUN
O:0
L
0
Bul.1762

CLAMP RETRACTED LIGHT
O:0
L
2
Bul.1762

FAULTY COMPONENT
O:0
U
5
Bul.1762

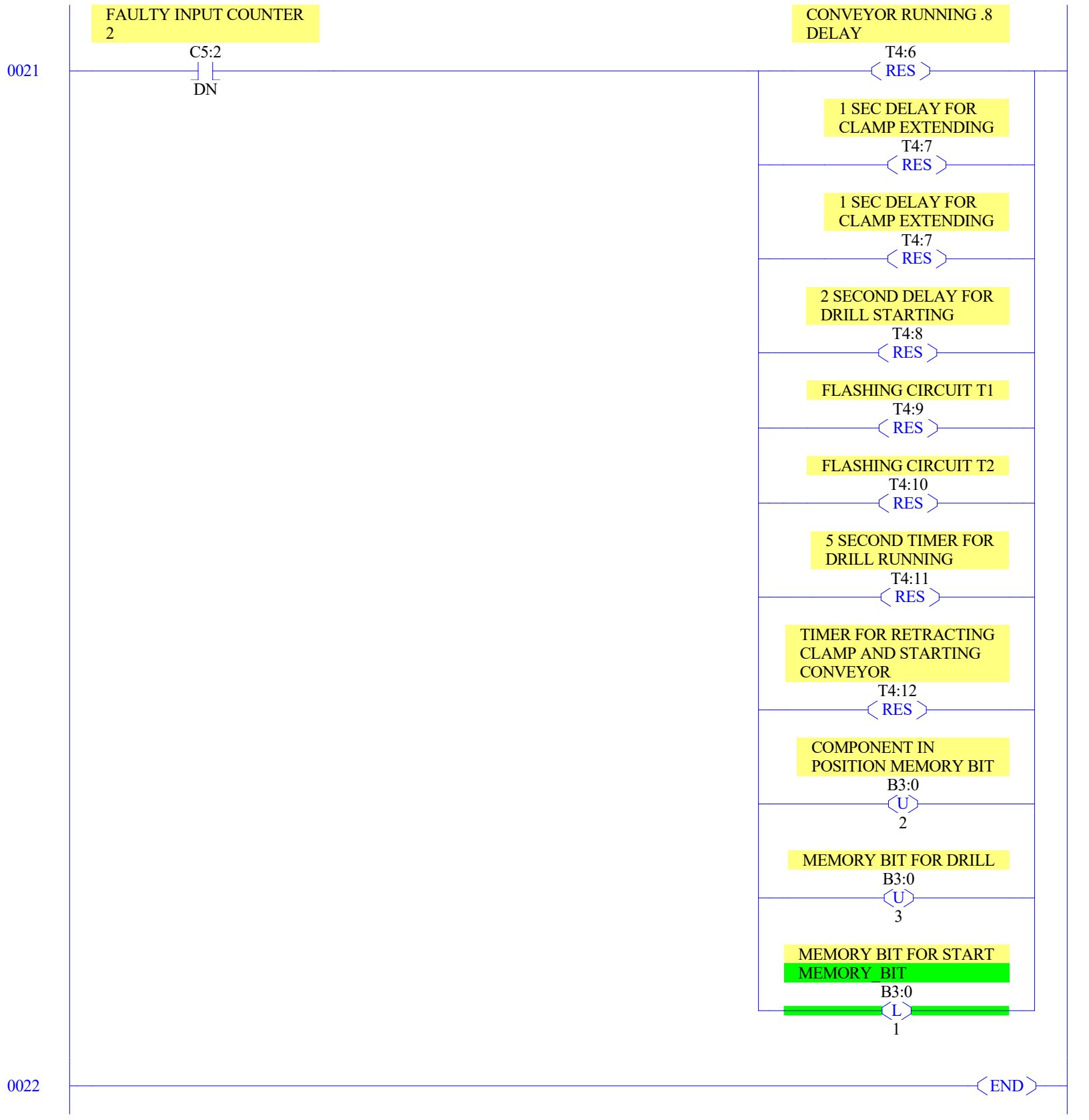
MEMORY BIT FOR LIGHT OUTPUTS
B3:0
U
5

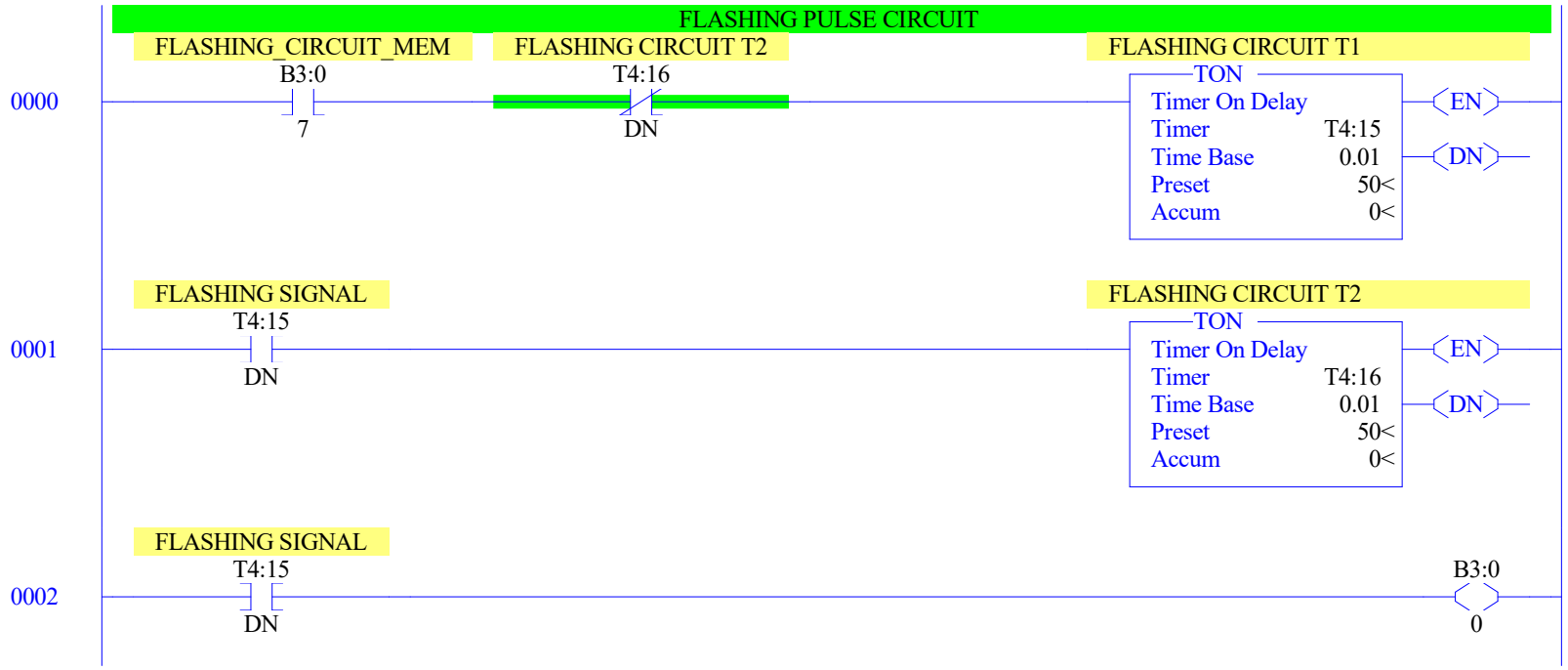
GOOD DETECTION RESTART CYCLE

0020

FAULTY COMPONENT
I:0
5
Bul.1762

FAULTY INPUT COUNTER
2
CTU
Count Up
Counter C5:2
Preset 2<
Accum 0<
CU
DN


FAULTY INPUT COUNTER 2
C5:2
DN
CONVEYOR RUNNING .8 DELAY
T4:6
RES
1 SEC DELAY FOR CLAMP EXTENDING
T4:7
RES
1 SEC DELAY FOR CLAMP EXTENDING
T4:7
RES
2 SECOND DELAY FOR DRILL STARTING
T4:8
RES
FLASHING CIRCUIT T1
T4:9
RES
FLASHING CIRCUIT T2
T4:10
RES
5 SECOND TIMER FOR DRILL RUNNING
T4:11
RES
TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR
T4:12
RES
COMPONENT IN POSITION MEMORY BIT
B3:0
U
2
MEMORY BIT FOR DRILL
B3:0
U
3
MEMORY BIT FOR START MEMORY_BIT
B3:0
L
1
0021
0022
END


FLASHING PULSE CIRCUIT
FLASHING_CIRCUIT_MEM
B3:0
7
FLASHING CIRCUIT T2
T4:16
DN
FLASHING CIRCUIT T1
TON
Timer On Delay
Timer T4:15
Time Base 0.01
Preset 50<
Accum 0<
EN
DN
0000
FLASHING SIGNAL
T4:15
DN
FLASHING CIRCUIT T2
TON
Timer On Delay
Timer T4:16
Time Base 0.01
Preset 50<
Accum 0<
EN
DN
0001
FLASHING SIGNAL
T4:15
DN
B3:0
0
0002

MANUAL CIRCUIT

0003

MANUAL BUTTON
I:0
1
Bul.1762

INPUT COUNTER FOR JOG
CTU
Count Up
Counter C5:3
Preset 6<
Accum 7<
CU
DN

CONVEYOR RUN JOG

0004

INPUT COUNTER FOR JOG
EQU
Equal
Source A C5:3.ACC 7<
Source B 1 1<

CONVEYOR RUN
O:0
0
Bul.1762

DRILL IN JOG

0005

INPUT COUNTER FOR JOG
EQU
Equal
Source A C5:3.ACC 7<
Source B 2 2<

MOV
Move
Source 0 0<
Dest O:0.0 0<

CLAMP RETRACT JOG

0006

INPUT COUNTER FOR JOG
EQU
Equal
Source A C5:3.ACC 7<
Source B 3 3<

CLAMP EXTEND INDICATION
O:0
3
Bul.1762

CLAMP EXTEND JOG

0007

INPUT COUNTER FOR JOG
EQU
Equal
Source A C5:3.ACC 7<
Source B 4 4<

DRILL INDICATION
O:0
1
Bul.1762

FLASHING_CIRCUIT_MEM
B3:0
L
7

BRING IN FLASHING CIRCUIT

0008

INPUT COUNTER FOR JOG

EQU
Equal
Source A C5:3.ACC
7<
Source B 5
5<

MOV
Move
Source 0
0<
Dest O:0.0
0<

FLASHING_CIRCUIT_MEM
B3:0
U
7

FLASHING INDICATION FOR DRILL RUNNING

0009

FLASHING SIGNAL
T4:15
DN

INDICATOR LIGHT
O:0
4
Bul.1762

UNLUATCH FLASHING CIRCUIT, BRING ON FAULTY COMPONENT

0010

INPUT COUNTER FOR JOG

EQU
Equal
Source A C5:3.ACC
7<
Source B 6
6<

CLAMP RETRACTED LIGHT
O:0
2
Bul.1762

RESET COUNTER

RESETS JOG COUNTER

INPUT COUNTER FOR JOG

INPUT COUNTER FOR JOG

0011

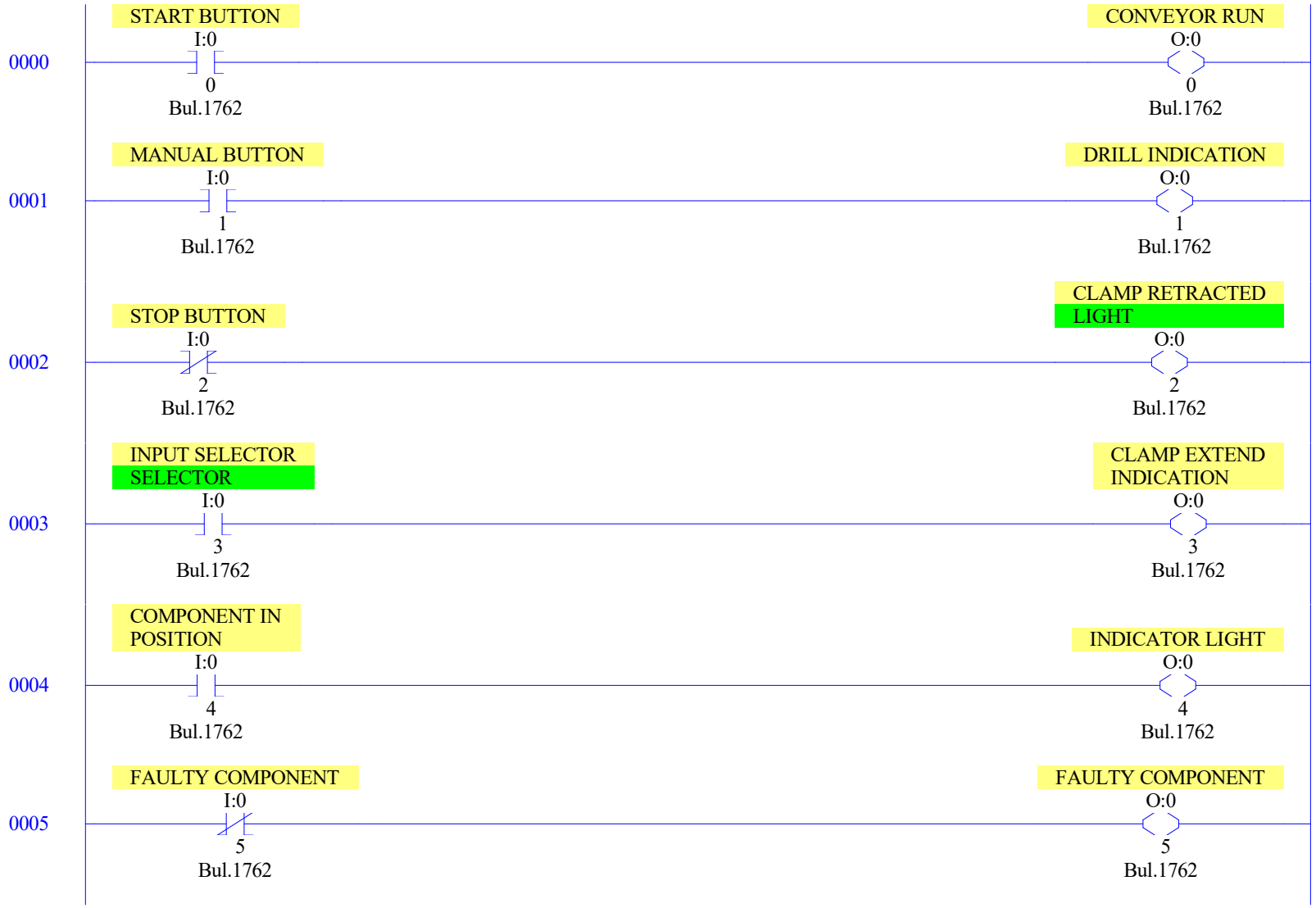
EQU
Equal
Source A C5:3.ACC
7<
Source B 8
8<

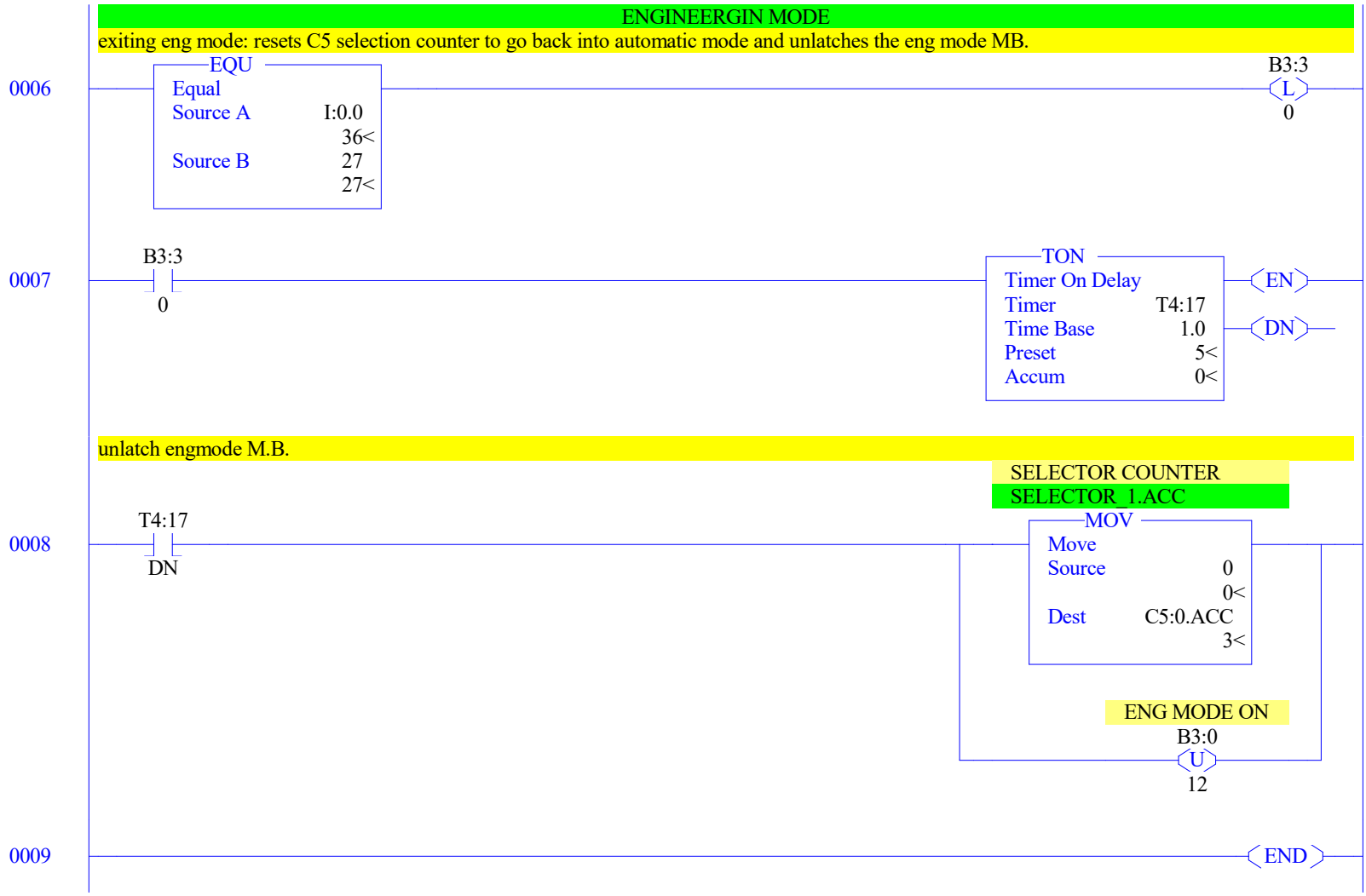
C5:3
RES

0012

END

LAD 5 - SUB2 - ENGINEER --- Total Rungs in File = 10




ENGINEERGIN MODE
exiting eng mode: resets C5 selection counter to go back into automatic mode and unlatches the eng mode MB.
0006
EQU
Equal
Source A I:0.0
36<
Source B 27
27<
B3:3
L
0
0007
B3:3
0
TON
Timer On Delay
Timer T4:17
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
unlatch engmode M.B.
SELECTOR COUNTER
SELECTOR_1.ACC
0008
T4:17
DN
MOV
Move
Source 0
0<
Dest C5:0.ACC
3<
ENG MODE ON
B3:0
U
12
0009
END

Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
O:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
I:0.0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1762	MicroLogix 1200 Series C (1 or 2

Main

Processor Mode S:1/0 - S:1/4 = Remote Run
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 1000-0011-0000-0100

Proc

OS Catalog Number S:57 = 1200
OS Series S:58 = C
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = A
Processor FRN S:62 =
User Program Type S:63 = 4110h
Compiler Revision Number S:64 =

Scan Times

Maximum (x10 ms) S:22 = 17
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 7
Scan Toggle Bit S:33/9 = 0

Math

Math Overflow Selected S:2/14 = 0
Overflow Trap S:5/0 = 0
Carry S:0/0 = 0
Overflow S:0/1 = 0
Zero Bit S:0/2 = 1
Sign Bit S:0/3 = 0
Math Register (lo word) S:13 = 0
Math Register (high word) S:14-S:13 = 0
Math Register (32 Bit) S:14-S:13 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = Remote Run
Node Address S:15 (low byte) = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0
Outgoing Msg Cmd Pending S:33/2 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0
Startup Protection Fault S:1/9 = 0
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0
Control Register Error S:5/2 = 0
Major Error Executing User
Fault Rtn. S:5/3 = 0
Retentive Data Lost S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
Fault Routine S:29 = 0
Major Error S:6 = 0h
Error Description:

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- BINARY

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B3:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	
B3:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
B3:4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol) Description
T4:0	0	0	0	1.0 sec	10	0	
T4:1	1	0	1	.01 sec	50	50	
T4:2	1	1	0	.01 sec	50	3	
T4:3	0	0	0	1.0 sec	10	0	
T4:4	0	0	0	1.0 sec	10	0	
T4:5	0	0	0	1.0 sec	10	0	
T4:6	0	0	0	.01 sec	80	0	CONVEYOR RUNNING .8 DELAY
T4:7	0	0	0	1.0 sec	1	0	1 SEC DELAY FOR CLAMP EXTENDING
T4:8	0	0	0	1.0 sec	2	0	2 SECOND DELAY FOR DRILL STARTING
T4:9	1	0	1	.01 sec	20	20	FLASHING CIRCUIT T1
T4:10	1	1	0	.01 sec	20	13	FLASHING CIRCUIT T2
T4:11	0	0	0	1.0 sec	5	0	5 SECOND TIMER FOR DRILL RUNNING
T4:12	0	0	0	1.0 sec	2	0	TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR
T4:13	0	0	0	1.0 sec	2	0	TIMER TO RESET COUNTER IF INPUTS TAKE LONGER THAN 2SEC
T4:14	0	0	0	1.0 sec	2	0	TIMER FOR FAULTY COMPONENT LIGHT
T4:15	0	0	0	.01 sec	50	0	FLASHING CIRCUIT T1
T4:16	0	0	0	.01 sec	50	0	FLASHING CIRCUIT T2
T4:17	0	0	0	1.0 sec	5	0	
T4:18	0	0	0	.01 sec	10	0	(INPUT_DELAY_TIMER)
T4:19	0	0	0	.01 sec	5	0	

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol) Description
C5:0	0	0	0	0	0	0	5	3	(SELECTOR 1) SELECTOR COUNTER
C5:1	0	0	0	0	0	0	4	0	FAULTY INPUT COUNTER1
C5:2	0	0	0	0	0	0	2	0	FAULTY INPUT COUNTER 2
C5:3	1	0	1	0	0	0	6	7	INPUT COUNTER FOR JOG

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol)	Description
R6:0	0	0	0	0	0	0	0	0	0	0		

Data File N7 (dec) -- INTEGER

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	0									

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0				

Address (Symbol) = Value [Description]

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
B3:0/1	MEMORY_BIT	Global	MEMORY BIT FOR START		
B3:0/2			COMPONENT IN POSITION MEMORY BIT		
B3:0/3			MEMORY BIT FOR DRILL		
B3:0/4			MEMORY BIT TO START TIMER FROM COUNTER INPUT		
B3:0/5			MEMORY BIT FOR LIGHT OUTPUTS		
B3:0/6			state A		
B3:0/7			FLASHING_CIRCUIT_MEM		
B3:0/8			state B		
B3:0/9			state C		
B3:0/10			PAUSE MEMORY		
B3:0/11			state D		
B3:0/12			ENG MODE ON		
B3:0/13			test		
B3:1/0					
C5:0	SELECTOR_1	Global	SELECTOR COUNTER		
C5:0.ACC					
C5:1			FAULTY INPUT COUNTER1		
C5:2			FAULTY INPUT COUNTER 2		
C5:3			INPUT COUNTER FOR JOG		
I:0.0					
I:0/0			START BUTTON		
I:0/1			MANUAL BUTTON		
I:0/2			STOP BUTTON		
I:0/3	SELECTOR	Global	INPUT SELECTOR		
I:0/4			COMPONENT IN POSITION		
I:0/5			FAULTY COMPONENT		
O:0.0					
O:0/0			CONVEYOR RUN		
O:0/1			DRILL INDICATION		
O:0/2	LIGHT	Global	CLAMP RETRACTED		
O:0/3			CLAMP EXTEND INDICATION		
O:0/4			INDICATOR LIGHT		
O:0/5			FAULTY COMPONENT		
S:0			Arithmetic Flags		
S:0/0			Processor Arithmetic Carry Flag		
S:0/1			Processor Arithmetic Underflow/ Overflow Flag		
S:0/2			Processor Arithmetic Zero Flag		
S:0/3			Processor Arithmetic Sign Flag		
S:1			Processor Mode Status/ Control		
S:1/0			Processor Mode Bit 0		
S:1/1			Processor Mode Bit 1		
S:1/2			Processor Mode Bit 2		
S:1/3			Processor Mode Bit 3		
S:1/4			Processor Mode Bit 4		
S:1/5			Forces Enabled		
S:1/6			Forces Present		
S:1/7			Comms Active		
S:1/8			Fault Override at Powerup		
S:1/9			Startup Protection Fault		
S:1/10			Load Memory Module on Memory Error		
S:1/11			Load Memory Module Always		
S:1/12			Load Memory Module and RUN		
S:1/13			Major Error Halted		
S:1/14			Access Denied		
S:1/15			First Pass		
S:2/0			STI Pending		
S:2/1			STI Enabled		
S:2/2			STI Executing		
S:2/3			Index Addressing File Range		
S:2/4			Saved with Debug Single Step		
S:2/5			DH-485 Incoming Command Pending		
S:2/6			DH-485 Message Reply Pending		
S:2/7			DH-485 Outgoing Message Command Pending		
S:2/15			Comms Servicing Selection		
S:3			Current Scan Time/ Watchdog Scan Time		
S:4			Time Base		
S:5/0			Overflow Trap		
S:5/2			Control Register Error		
S:5/3			Major Err Detected Executing UserFault Routine		
S:5/4			M0-M1 Referenced on Disabled Slot		
S:5/8			Memory Module Boot		
S:5/9			Memory Module Password Mismatch		
S:5/10			STI Overflow		
S:5/11			Battery Low		
S:6			Major Error Fault Code		
S:7			Suspend Code		
S:8			Suspend File		
S:9			Active Nodes		
S:10			Active Nodes		
S:11			I/O Slot Enables		
S:12			I/O Slot Enables		
S:13			Math Register		
S:14			Math Register		
S:15			Node Address/ Baud Rate		

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
S:16			Debug Single Step Rung		
S:17			Debug Single Step File		
S:18			Debug Single Step Breakpoint Rung		
S:19			Debug Single Step Breakpoint File		
S:20			Debug Fault/ Powerdown Rung		
S:21			Debug Fault/ Powerdown File		
S:22			Maximum Observed Scan Time		
S:23			Average Scan Time		
S:24			Index Register		
S:25			I/O Interrupt Pending		
S:26			I/O Interrupt Pending		
S:27			I/O Interrupt Enabled		
S:28			I/O Interrupt Enabled		
S:29			User Fault Routine File Number		
S:30			STI Setpoint		
S:31			STI File Number		
S:32			I/O Interrupt Executing		
S:33			Extended Proc Status Control Word		
S:33/0			Incoming Command Pending		
S:33/1			Message Reply Pending		
S:33/2			Outgoing Message Command Pending		
S:33/3			Selection Status User/DF1		
S:33/4			Communicat Active		
S:33/5			Communicat Servicing Selection		
S:33/6			Message Servicing Selection Channel 0		
S:33/7			Message Servicing Selection Channel 1		
S:33/8			Interrupt Latency Control Flag		
S:33/9			Scan Toggle Flag		
S:33/10			Discrete Input Interrupt Reconfigur Flag		
S:33/11			Online Edit Status		
S:33/12			Online Edit Status		
S:33/13			Scan Time Timebase Selection		
S:33/14			DTR Control Bit		
S:33/15			DTR Force Bit		
S:34			Pass-thru Disabled		
S:34/0			Pass-Thru Disabled Flag		
S:34/1			DH+ Active Node Table Enable Flag		
S:34/2			Floating Point Math Flag Disable,Fl		
S:35			Last 1 ms Scan Time		
S:36			Extended Minor Error Bits		
S:36/8			DII Lost		
S:36/9			STI Lost		
S:36/10			Memory Module Data File Overwrite Protection		
S:37			Clock Calendar Year		
S:38			Clock Calendar Month		
S:39			Clock Calendar Day		
S:40			Clock Calendar Hours		
S:41			Clock Calendar Minutes		
S:42			Clock Calendar Seconds		
S:43			STI Interrupt Time		
S:44			I/O Event Interrupt Time		
S:45			DII Interrupt Time		
S:46			Discrete Input Interrupt- File Number		
S:47			Discrete Input Interrupt- Slot Number		
S:48			Discrete Input Interrupt- Bit Mask		
S:49			Discrete Input Interrupt- Compare Value		
S:50			Processor Catalog Number		
S:51			Discrete Input Interrupt- Return Number		
S:52			Discrete Input Interrupt- Accumulat		
S:53			Reserved/ Clock Calendar Day of the Week		
S:55			Last DII Scan Time		
S:56			Maximum Observed DII Scan Time		
S:57			Operating System Catalog Number		
S:58			Operating System Series		
S:59			Operating System FRN		
S:61			Processor Series		
S:62			Processor Revision		
S:63			User Program Type		
S:64			User Program Functional Index		
S:65			User RAM Size		
S:66			Flash EEPROM Size		
S:67			Channel 0 Active Nodes		
S:68			Channel 0 Active Nodes		
S:69			Channel 0 Active Nodes		
S:70			Channel 0 Active Nodes		
S:71			Channel 0 Active Nodes		
S:72			Channel 0 Active Nodes		
S:73			Channel 0 Active Nodes		
S:74			Channel 0 Active Nodes		
S:75			Channel 0 Active Nodes		
S:76			Channel 0 Active Nodes		
S:77			Channel 0 Active Nodes		
S:78			Channel 0 Active Nodes		
S:79			Channel 0 Active Nodes		

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code
S:80			Channel 0 Active Nodes		
S:81			Channel 0 Active Nodes		
S:82			Channel 0 Active Nodes		
S:83			DH+ Active Nodes		
S:84			DH+ Active Nodes		
S:85			DH+ Active Nodes		
S:86			DH+ Active Nodes		
T4:0					
T4:0/DN					
T4:1/DN					
T4:3					
T4:3/DN					
T4:4					
T4:4/DN					
T4:5					
T4:5/DN					
T4:6			CONVEYOR RUNNING .8 DELAY		
T4:7			1 SEC DELAY FOR CLAMP EXTENDING		
T4:8			2 SECOND DELAY FOR DRILL STARTING		
T4:8/DN					
T4:9			FLASHING CIRCUIT T1		
T4:9/DN					
T4:10			FLASHING CIRCUIT T2		
T4:10/DN					
T4:11			5 SECOND TIMER FOR DRILL RUNNING		
T4:12			TIMER FOR RETRACTING CLAMP AND STARTING CONVEYOR		
T4:13			TIMER TO RESET COUNTER IF INPUTS TAKE LONGER THAN 2SEC		
T4:14			TIMER FOR FAULTY COMPONENT LIGHT		
T4:15			FLASHING CIRCUIT T1		
T4:15/DN			FLASHING SIGNAL		
T4:16			FLASHING CIRCUIT T2		
T4:17/DN					
T4:18	INPUT_DELAY_TIMER	Global			
T4:18/DN					
T4:19/DN					
U:3			sub0		
U:4			sub1		
U:5			sub2		

Instruction Comment Database

Address	Instruction	Description

Symbol Group Database

Group_Name Description